WEEK 7: SQL QUERY IMPLEMENTATION FOR ARTHUB

1. Objective

The main objective of this exercise is to implement SQL queries in the ArtHub database to retrieve, filter, sort, and analyze customer, artwork, order, payment, and review information. It also aims to generate basic business reports that help manage artwork availability, customer details, orders, and product performance.

2. Scenario

ArtHub is an online art marketplace that allows customers to explore and purchase artworks created by artists. The database stores information about customers, artworks, orders, payments, and reviews.

The management team requires useful information from the database for daily operations and basic business analysis. SQL queries are used to retrieve the required details, apply filtering conditions, sort records, identify unique values, and generate reports.

3. Perform Basic SQL Queries

Basic SQL queries are used to retrieve information from the ArtHub database. The SELECT statement helps display records stored in different tables.

The following operations were performed:

- Displayed all customer details.

```
SQL> SELECT * FROM Customer;

CUSTOMER_ID CUSTOMER_NAME                  EMAIL
----------- ------------------------------ ------------------------------
PHONE                          ADDRESS                        REGISTRATION_DATE
------------------------------ ------------------------------ ------------------
          1 Dhoni                          Dhoni77@gmail.com
9876543210                     Chennai                        21-APR-21

          2 Virat                          Virat18@gmail.com
9876543123                     Bangalore                      24-JUN-22

          3 Ashwin                         ashwin99@gmail.com
9876543212                     Chennai                        11-SEP-21


CUSTOMER_ID CUSTOMER_NAME                  EMAIL
----------- ------------------------------ ------------------------------
PHONE                          ADDRESS                        REGISTRATION_DATE
------------------------------ ------------------------------ ------------------
          4 SanjuSamson                    sanju11@gmail.com
9876543234                     Kerala                         11-SEP-21

          5 Shreyas Iyer                   shreyas96@gmail.com
9876545689                     Hydrebad                       27-MAR-25

          6 Hardhik Pandya                 hardhik33@gmail.com
9876598765                     Hyderabad                      30-MAR-19


CUSTOMER_ID CUSTOMER_NAME                  EMAIL
----------- ------------------------------ ------------------------------
PHONE                          ADDRESS                        REGISTRATION_DATE
------------------------------ ------------------------------ ------------------
          7 Ruturaj Gaikwad                ruturaj31@gmail.com
983214561                      Chennai                        20-MAR-20

          8 Rohit Sharma                   rohit45@gmail.com
9876512340                     Bangalore                      14-SEP-18
```

- Retrieved artworks available in stock.

```
SQL> SELECT * FROM ArtProduct WHERE Stock > 0;

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
101        1          1
Sunset Over Hills
      5000          3

102        2          1
Colors of Nature
      5200          2

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

103        3          2
Pencil Portrait
      1800          5

104        4          2
Charcoal Dreams

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
      2200          4

105        5          3
Digital Galaxy
```

```
106        6          3

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
Future City
      4000          3

107        7          4
Morning Light
      2800          4


ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
108        8          5
The Thinker Sculpture
      7500          2

109        9          9
Abstract Dreams
      4200          3

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

110        10         10
Handmade Craft
      2500          7
```

- Displayed artwork names and their prices.

```
SQL> SELECT Artwork_Name, Price FROM ArtProduct;

ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Sunset Over Hills
      5000

Colors of Nature
      5200

Pencil Portrait
      1800


ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Charcoal Dreams
      2200

Digital Galaxy
      3500

Future City
      4000


ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Morning Light
      2800

The Thinker Sculpture
      7500
```

- Retrieved all customer orders.

```
SQL> SELECT * FROM Orders;

ORDER_ID   CUSTOMER_ID ORDER_DATE         TOTAL_AMOUNT ORDER_STATUS
---------- ----------- ------------------ ------------ --------------------
O1                   1 20-SEP-26                  5000 Delivered
O2                   2 21-SEP-26                  5200 Shipped
O3                   3 21-SEP-26                  1800 Delivered
O4                   4 22-SEP-26                  2200 Processing
O5                   5 22-SEP-26                  3500 Shipped
O6                   6 23-SEP-26                  4000 Delivered
O7                   7 24-SEP-26                  2800 Processing

7 rows selected.
```

- Displayed payment details.

```
SQL> SELECT * FROM Payment;

PAYMENT_ID ORDER_ID   PAYMENT_DATE        PAYMENT_MODE
---------- ---------- ------------------- --------------------
PAYMENT_STATUS        TRANSACTION_AMOUNT
--------------------- ------------------
P1         O1          20-SEP-26          UPI
Completed                             5000

P2         O2          21-SEP-26          Card
Completed                             5200

P3         O3          21-SEP-26          UPI
Completed                             1800


PAYMENT_ID ORDER_ID   PAYMENT_DATE        PAYMENT_MODE
---------- ---------- ------------------- --------------------
PAYMENT_STATUS        TRANSACTION_AMOUNT
--------------------- ------------------
P4         O4          22-SEP-26          Net Banking
Pending                               2200

P6         O6          23-SEP-26          UPI
Completed                             4000

P5         O5          22-SEP-26          Card
Completed                             3500


PAYMENT_ID ORDER_ID   PAYMENT_DATE        PAYMENT_MODE
---------- ---------- ------------------- --------------------
PAYMENT_STATUS        TRANSACTION_AMOUNT
--------------------- ------------------
P7         O7          24-SEP-26          Card
Pending                               2800


7 rows selected.
```

These operations helped verify the data stored in the database and understand the information available in each table.

4. Apply Filtering Conditions Using WHERE

The WHERE clause is used to retrieve records that satisfy specific conditions. It helps identify the required information without displaying unrelated records.

The following filtering operations were performed:

- Identified artworks priced above ₹5,000.

```
SQL> SELECT * FROM ArtProduct WHERE Price > 5000;

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
102        2          1
Colors of Nature
      5200          2

108        8          5
The Thinker Sculpture
      7500          2

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
```

- Retrieved artworks available in stock.

```
SQL> SELECT * FROM ArtProduct WHERE Stock > 0;

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
101        1          1
Sunset Over Hills
      5000          3

102        2          1
Colors of Nature
      5200          2

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

103        3          2
Pencil Portrait
      1800          5

104        4          2
Charcoal Dreams

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
      2200          4

105        5          3
```

```
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
Future City
      4000          3

107        7          4
Morning Light
      2800          4


ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
108        8          5
The Thinker Sculpture
      7500          2

109        9          9
Abstract Dreams
      4200          3

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

110        10         10
Handmade Craft
      2500          7


10 rows selected.
```

- Displayed delivered orders based on the order statuses available in ArtHub.

```
SQL> SELECT * FROM Orders WHERE order_Status = 'Delivered';

ORDER_ID   CUSTOMER_ID ORDER_DATE          TOTAL_AMOUNT ORDER_STATUS
---------- ----------- ------------------ ------------ --------------------
O1                   1 20-SEP-26                  5000 Delivered
O3                   3 21-SEP-26                  1800 Delivered
O6                   6 23-SEP-26                  4000 Delivered
```

- Retrieved reviews with ratings greater than 4.

```
SQL> SELECT * FROM Review WHERE Rating > 4;

REVIEW_ID  CUSTOMER_ID ARTWORK_ID REVIEW_DATE           RATING
---------- ----------- ---------- ------------------ ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
R1                   1 101        25-SEP-26                   5
Beautiful artwork with excellent colors

R4                   4 104        28-SEP-26                   5
Excellent charcoal artwork with fine details

R7                   7 107        01-OCT-26                   5
Beautiful painting with a peaceful atmosphere


REVIEW_ID  CUSTOMER_ID ARTWORK_ID REVIEW_DATE           RATING
---------- ----------- ---------- ------------------ ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
R10                 10 110        04-OCT-26                   5
Beautiful handmade craft with excellent finishing
```

These queries helped filter artwork, order, and review information according to the given conditions.

5. Sort Data Using ORDER BY

The ORDER BY clause is used to arrange records in ascending or descending order. It makes the retrieved information easier to read and compare.

The following sorting operations were performed:

- Arranged artworks from the lowest to the highest price.

```
SQL> SELECT * FROM ArtProduct ORDER BY Price ASC;

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
103        3          2
Pencil Portrait
      1800          5

104        4          2
Charcoal Dreams
      2200          4

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

110        10         10
Handmade Craft
      2500          7

107        7          4
Morning Light

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
      2800          4

105        5          3
```

- Displayed customers in alphabetical order.

```
SQL> SELECT * FROM Customer ORDER BY Customer_Name ASC;

CUSTOMER_ID CUSTOMER_NAME                  EMAIL
----------- ------------------------------ ------------------------------
PHONE                          ADDRESS                        REGISTRATION_DATE
------------------------------ ------------------------------ ------------------
          3 Ashwin                         ashwin99@gmail.com
9876543212                     Chennai                        11-SEP-21

          1 Dhoni                          Dhoni77@gmail.com
9876543210                     Chennai                        21-APR-21

          6 Hardhik Pandya                 hardhik33@gmail.com
9876598765                     Hyderabad                      30-MAR-19


CUSTOMER_ID CUSTOMER_NAME                  EMAIL
----------- ------------------------------ ------------------------------
PHONE                          ADDRESS                        REGISTRATION_DATE
------------------------------ ------------------------------ ------------------
         10 Jadeja                         jadeja8@gmail.com
9123456781                     Chennai                        12-JUL-21

          9 Raina                          raina3@gmail.com
9812345390                     Chennai                        12-NOV-19

          8 Rohit Sharma                   rohit45@gmail.com
9876512340                     Bangalore                      14-SEP-18


CUSTOMER_ID CUSTOMER_NAME                  EMAIL
----------- ------------------------------ ------------------------------
PHONE                          ADDRESS                        REGISTRATION_DATE
------------------------------ ------------------------------ ------------------
          7 Ruturaj Gaikwad                ruturaj31@gmail.com
983214561                      Chennai                        20-MAR-20

          4 SanjuSamson                    sanju11@gmail.com
9876543234                     Kerala                         11-SEP-21
```

- Identified the most expensive artworks by arranging prices in descending order.

```
SQL> SELECT * FROM ArtProduct ORDER BY Price DESC;

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
108        8          5
The Thinker Sculpture
      7500          2

102        2          1
Colors of Nature
      5200          2

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

101        1          1
Sunset Over Hills
      5000          3

109        9          9
Abstract Dreams

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
      4200          3

106        6          3
Future City
```

- Displayed the latest orders first according to their order dates.

```
SQL> SELECT * FROM Orders ORDER BY Order_Date DESC;

ORDER_ID   CUSTOMER_ID ORDER_DATE         TOTAL_AMOUNT ORDER_STATUS
---------- ----------- ------------------ ------------ --------------------
07                   7 24-SEP-26                  2800 Processing
06                   6 23-SEP-26                  4000 Delivered
04                   4 22-SEP-26                  2200 Processing
05                   5 22-SEP-26                  3500 Shipped
02                   2 21-SEP-26                  5200 Shipped
03                   3 21-SEP-26                  1800 Delivered
01                   1 20-SEP-26                  5000 Delivered

7 rows selected.
```

These operations helped organize the database records for easier analysis.

6. Retrieve Unique Values Using DISTINCT

The DISTINCT keyword is used to retrieve unique values and eliminate duplicate entries from the result.

The following operations were performed:

- Retrieved the unique artwork category IDs available in ArtHub.

```
SQL> SELECT DISTINCT Category_ID FROM ArtProduct;

CATEGORY_I
----------
1
3
10
5
9
2
4

7 rows selected.
```

- Identified the different payment methods used by customers.

```
SQL> SELECT DISTINCT Payment_Mode FROM Payment;

PAYMENT_MODE
--------------------
Net Banking
Card
UPI
```

These operations helped identify the categories and payment methods represented in the database.

7. Search Artworks Based on Conditions

SQL filtering conditions can be used to search for artworks according to price, category, availability, name, and stock quantity.

The following operations were performed:

- Retrieved artworks within a suitable price range based on the prices available in ArtHub.

```
SQL> SELECT * FROM ArtProduct WHERE Price BETWEEN 2000 AND 5000;

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
101        1          1
Sunset Over Hills
      5000          3

104        4          2
Charcoal Dreams
      2200          4

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

105        5          3
Digital Galaxy
      3500          6

106        6          3
Future City

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
      4000          3

107        7          4
Morning Light
```

- Displayed artworks belonging to a particular category.

```
SQL> SELECT * FROM ArtProduct WHERE Category_ID = '1';

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
101        1          1
Sunset Over Hills
      5000          3

102        2          1
Colors of Nature
      5200          2

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
```

- Identified artworks currently available in inventory.

```
SQL> SELECT * FROM ArtProduct WHERE Stock > 0;

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
101        1          1
Sunset Over Hills
      5000          3

102        2          1
Colors of Nature
      5200          2

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

103        3          2
Pencil Portrait
      1800          5

104        4          2
Charcoal Dreams

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
      2200          4

105        5          3
```

```
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
Future City
      4000          3

107        7           4
Morning Light
      2800          4


ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
108        8           5
The Thinker Sculpture
      7500          2

109        9           9
Abstract Dreams
      4200          3

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

110        10          10
Handmade Craft
      2500          7


10 rows selected.
```

- Searched for artworks using keywords in their names.

```
SQL> SELECT * FROM ArtProduct WHERE Artwork_Name LIKE '%Dreams%';

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
104        4           2
Charcoal Dreams
      2200          4

109        9           9
Abstract Dreams
      4200          3

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
```

- Identified artworks with low stock quantities.

```
SQL> SELECT * FROM ArtProduct WHERE Stock < 3;

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
102        2           1
Colors of Nature
      5200          2

108        8           5
The Thinker Sculpture
      7500          2

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
```

These operations helped retrieve artworks based on specific requirements and supported inventory management.

8. Retrieve Customer and Artwork Information

Customer and artwork information can be retrieved together to understand purchasing activities. Relationships between the Customer, Orders, OrderDetails, and ArtProduct tables were used where required.

The following operations were performed:

- Displayed customer details along with their orders.

```
SQL> SELECT Customer.Customer_ID, Customer.Customer_Name, Orders.Order_ID, Orders.Total_Amount, Orders.Order_Status FROM Customer JOIN Orders ON Customer.Cu
stomer_ID = Orders.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                  ORDER_ID   TOTAL_AMOUNT
----------- ------------------------------ ---------- ------------
ORDER_STATUS
--------------------
          1 Dhoni                          O1                5000
Delivered

          2 Virat                          O2                5200
Shipped

          3 Ashwin                         O3                1800
Delivered


CUSTOMER_ID CUSTOMER_NAME                  ORDER_ID   TOTAL_AMOUNT
----------- ------------------------------ ---------- ------------
ORDER_STATUS
--------------------
          4 SanjuSamson                    O4                2200
Processing

          5 Shreyas Iyer                   O5                3500
Shipped

          6 Hardhik Pandya                 O6                4000
Delivered


CUSTOMER_ID CUSTOMER_NAME                  ORDER_ID   TOTAL_AMOUNT
----------- ------------------------------ ---------- ------------
ORDER_STATUS
--------------------
          7 Ruturaj Gaikwad                O7                2800
Processing
```

- Retrieved artwork details, including artwork names, prices, and category IDs.

```
SQL> SELECT Artwork_ID, Artwork_Name, Price, Category_ID FROM ArtProduct;

ARTWORK_ID
----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_I
---------- ----------
101
Sunset Over Hills
      5000 1

102
Colors of Nature
      5200 1

ARTWORK_ID
----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_I
---------- ----------

103
Pencil Portrait
      1800 2

104
Charcoal Dreams

ARTWORK_ID
----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_I
---------- ----------
      2200 2

105
```

- Identified customers who purchased a specific artwork.

```
SQL> SELECT Customer.Customer_ID, Customer.Customer_Name, Orders.Order_ID, OrderDetails.Artwork_ID FROM Customer JOIN Orders ON Customer.Customer_ID = Order
s.Customer_ID JOIN OrderDetails ON Orders.Order_ID = OrderDetails.Order_ID WHERE OrderDetails.Artwork_ID = '101';

CUSTOMER_ID CUSTOMER_NAME                ORDER_ID   ARTWORK_ID
----------- ---------------------------- ---------- ----------
          1 Dhoni                        01         101
```

- Displayed artworks purchased by customers.

```
SQL> SELECT Customer.Customer_Name, OrderDetails.Artwork_ID, ArtProduct.Artwork_Name FROM Customer JOIN Orders ON Customer.Customer_ID = Orders.Customer_ID
JOIN OrderDetails ON Orders.Order_ID = OrderDetails.Order_ID JOIN ArtProduct ON OrderDetails.Artwork_ID = ArtProduct.Artwork_ID;

CUSTOMER_NAME                ARTWORK_ID
---------------------------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
Dhoni                        101
Sunset Over Hills

Virat                        102
Colors of Nature

Ashwin                       103
Pencil Portrait


CUSTOMER_NAME                ARTWORK_ID
---------------------------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
SanjuSamson                  104
Charcoal Dreams

Shreyas Iyer                 105
Digital Galaxy

Hardhik Pandya               106
Future City


6 rows selected.
```

These queries helped connect customer information with order and artwork records. The number of matching results depended on the records available in the related tables.

9. Apply Multiple Filtering Conditions

Multiple filtering conditions allow records to be retrieved by applying more than one requirement at a time.

The following operations were performed:

- Filtered artworks using category and price conditions supported by the ArtHub database.

```
SQL> SELECT * FROM ArtProduct WHERE Category_ID = '3' AND Price > 3000;

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
105        5          3
Digital Galaxy
      3500          6

106        6          3
Future City
      4000          3

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
```

- Retrieved orders placed within a specified date range.

```
SQL> SELECT * FROM Orders WHERE Order_Date BETWEEN TO_DATE('20-09-2026','DD-MM-YYYY') AND TO_DATE('24-09-2026','DD-MM-YYYY');

ORDER_ID   CUSTOMER_ID ORDER_DATE         TOTAL_AMOUNT ORDER_STATUS
--------- ----------- ----------------- ------------ --------------------
01                   1 20-SEP-26                 5000 Delivered
02                   2 21-SEP-26                 5200 Shipped
03                   3 21-SEP-26                 1800 Delivered
04                   4 22-SEP-26                 2200 Processing
05                   5 22-SEP-26                 3500 Shipped
06                   6 23-SEP-26                 4000 Delivered
07                   7 24-SEP-26                 2800 Processing

7 rows selected.
```

10. Generate Basic Business Reports

10.1 Product Availability Report

The product availability report provides information about the artworks stored in ArtHub. It includes artwork names, prices, stock quantities, and availability status.

The report showed that the listed artworks had stock available at the time of execution. This information helps monitor inventory and identify artworks that may require restocking in the future.

10.2 Customer Report

The customer report provides information about the total number of registered customers and new customer registrations.

The total customer count was retrieved successfully. The search for registrations during the current month returned no records because no customer registration dates matched that condition. The registration-date details were also examined to understand the available customer data.

10.3 Order Report

The order report summarizes order information and helps monitor order processing.

The following details were examined:

- Total number of orders.

```
SQL> SELECT COUNT(*) AS Total_Orders FROM Orders;

TOTAL_ORDERS
------------
           7
```

- Number of delivered orders.

```
SQL> SELECT COUNT(*) AS Completed_Orders FROM Orders WHERE Order_Status = 'Delivered';

COMPLETED_ORDERS
----------------
               3
```

- Number of pending orders.

```
SQL> SELECT COUNT(*) AS Pending_Orders FROM Orders WHERE Order_Status = 'Pending';

PENDING_ORDERS
--------------
             0
```

- Number of cancelled orders.

```
SQL> SELECT COUNT(*) AS Cancelled_Orders FROM Orders WHERE Order_Status = 'Cancelled';

CANCELLED_ORDERS
----------------
               0
```

- Distribution of orders according to their recorded status.

```
SQL> SELECT Order_Status, COUNT(*) AS Total_Orders FROM Orders GROUP BY Order_Status;

ORDER_STATUS         TOTAL_ORDERS
-------------------- ------------
Processing                      2
Delivered                       3
Shipped                         2
```

The ArtHub database contains the statuses Delivered, Shipped, and Processing. Therefore, the report was adapted to the statuses actually stored in the Orders table. Pending and Cancelled counts returned zero because those status values were not present in the available order records.

10.4 Product Performance Report

The product performance report helps evaluate artwork pricing, customer feedback, and availability.

The following operations were performed:

- Identified the highest-priced artwork.

```
SQL> SELECT Artwork_Name, Price FROM ArtProduct WHERE Price = (SELECT MAX(Price) FROM ArtProduct);

ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE
----------
The Thinker Sculpture
      7500
```

- Counted the reviews associated with each artwork.

```
SQL> SELECT * FROM Review WHERE Rating > 4;

REVIEW_ID  CUSTOMER_ID ARTWORK_ID REVIEW_DATE            RATING
---------- ----------- ---------- ------------------ ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
R1                   1 101        25-SEP-26                   5
Beautiful artwork with excellent colors

R4                   4 104        28-SEP-26                   5
Excellent charcoal artwork with fine details

R7                   7 107        01-OCT-26                   5
Beautiful painting with a peaceful atmosphere


REVIEW_ID  CUSTOMER_ID ARTWORK_ID REVIEW_DATE            RATING
---------- ----------- ---------- ------------------ ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
R10                 10 110        04-OCT-26                   5
Beautiful handmade craft with excellent finishing
```

- Retrieved the average ratings for artworks.

```
SQL> SELECT Artwork_ID, AVG(Rating) AS Average_Rating FROM Review GROUP BY Artwork_ID HAVING AVG(Rating) < 3;

ARTWORK_ID AVERAGE_RATING
---------- --------------
106                   2
```

- Identified artworks available in stock.

```
SQL> SELECT * FROM ArtProduct WHERE Stock > 0;

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
101        1          1
Sunset Over Hills
      5000          3

102        2          1
Colors of Nature
      5200          2

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

103        3          2
Pencil Portrait
      1800          5

104        4          2
Charcoal Dreams

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
      2200          4

105        5          3
```

```
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
Future City
      4000          3

107        7          4
Morning Light
      2800          4


ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
108        8          5
The Thinker Sculpture
      7500          2

109        9          9
Abstract Dreams
      4200          3

ARTWORK_ID ARTIST_ID  CATEGORY_I
---------- ---------- ----------
ARTWORK_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

110        10         10
Handmade Craft
      2500          7


10 rows selected.
```

The Thinker Sculpture was identified as the highest-priced artwork at ₹7,500. The review analysis showed that each of the ten artworks had one recorded review. The report also helped identify the ratings and availability of individual artworks.

11. Result

The SQL query implementation for the ArtHub database was completed successfully. Basic data retrieval, filtering, sorting, unique-value selection, artwork searches, customer-order retrieval, multiple-condition filtering, and business report generation were performed.

The queries were adapted to the existing ArtHub table names, column names, price ranges, and order statuses. Some queries returned no matching records because the required values were not available in the database. These results reflected the existing data and did not necessarily indicate an SQL error.

12. Conclusion

This exercise provided practical experience in using SQL to retrieve and analyze data in the ArtHub database. The implementation demonstrated the use of SELECT, WHERE, ORDER BY, DISTINCT, filtering conditions, table relationships, aggregate functions, and business reporting operations.

The exercise also helped develop an understanding of how database structure and available records influence query results. Overall, the implementation improved the ability to manage customer, artwork, order, payment, and review information using SQL.